

NPM and package.json in Node.js

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① Title

NPM and package.json in Node.js

② Aim

To understand NPM and package.json, install an external package and use the package in a Node.js application.

③ Objective

- To understand NPM and its role in Node.js projects.
- To understand the purpose of the package.json file.
- To learn how dependencies are installed and managed.
- To install and use the external lodash package.
- To execute and verify a Node.js application using the installed package.

④ Theory

NPM (Node Package Manager) is the package manager used with Node.js to install, update and manage JavaScript packages and project dependencies.

package.json is the configuration file of a Node.js project. It stores the project name, version, main file, scripts and dependencies.

External package: In this experiment Lodash is installed as an external dependency. Lodash provides utility functions for common JavaScript operations.

Step	NPM workflow
1	Create the Node.js project folder
2	<code>npm init -y</code> creates package.json
3	<code>npm install lodash</code> installs the package
4	Import lodash in app.js
5	Use the lodash functions and run the application

⑤ Algorithm / Procedure

1. Create a new Node.js project folder.
2. Open the project folder in the terminal.
3. Initialize the project using `npm init -y`.
4. Install the external Lodash package using `npm install lodash`.
5. Verify that Lodash is added to the project dependencies.
6. Create `app.js` and import the Lodash package.
7. Create an array of numbers.
8. Use Lodash functions to calculate the sum, maximum and minimum values.
9. Run the application using `node app.js`.
10. Verify the actual output in the terminal.

Important NPM commands

Command	Purpose
npm init -y	Creates package.json with default values (the -y option accepts all defaults).
npm install lodash	Downloads Lodash into node_modules and adds it to the dependencies in package.json.
npm install	Installs every dependency listed in package.json (used after cloning a project).
node app.js	Runs the application.

6 Program

Project structure

```
Q7-NPM-and-package-json/  
├─ app.js  
├─ package.json  
├─ package-lock.json  
└─ node_modules/      (generated by NPM – not uploaded to GitHub)
```

app.js

```
// Experiment 6-7: using the external lodash package  
const _ = require("lodash");  
  
// Numbers can be typed after the file name (node app.js 5 9 2);  
// otherwise a default list is used.  
const typed = process.argv.slice(2).map(Number);  
const numbers = typed.length > 0 ? typed : [12, 7, 45, 3, 28, 19];  
  
if (numbers.some(Number.isNaN)) {  
  console.log("Invalid input: please enter numbers only.");  
  process.exit(1);  
}  
  
// Lodash utility functions  
const total = _.sum(numbers);  
const largest = _.max(numbers);  
const smallest = _.min(numbers);  
const average = _.mean(numbers);  
  
console.log("Numbers   :", numbers);  
console.log("Sum       :", total);  
console.log("Maximum   :", largest);  
console.log("Minimum   :", smallest);  
console.log("Average   :", average);
```

package.json (after npm install lodash)

```
{  
  "name": "q7-npm-and-package-json",  
  "version": "1.0.0",  
  "description": "Using the lodash package from NPM",  
  "main": "app.js",  
  "scripts": {  
    "start": "node app.js"  
  },  
  "license": "ISC",  
  "dependencies": {  
    "lodash": "^4.17.21"  
  }  
}
```

7 Code Explanation

Code / Method	Explanation
<code>const _ = require("lodash")</code>	Imports the installed Lodash package. Lodash is found in the <code>node_modules</code> folder created by NPM.
<code>process.argv.slice(2).map(Number)</code>	Reads any numbers typed after <code>node app.js</code> and converts them from text to numbers.
<code>const numbers = typed.length > 0 ? typed : [12, 7, 45, 3, 28, 19]</code>	Uses the typed numbers if there are any; otherwise uses the default list of six numbers.
<code>numbers.some(Number.isNaN)</code>	Checks whether any input is not a number. If so, an error message is printed and the program stops.
<code>_.sum(numbers)</code>	Lodash function that adds all the values of the array.
<code>_.max(numbers), _.min(numbers)</code>	Lodash functions that return the largest and the smallest value of the array.
<code>_.mean(numbers)</code>	Lodash function that returns the average (sum divided by the count).
<code>console.log()</code>	Displays the array and the calculated results in the terminal.

8 Test Cases and Execution Data

The NPM commands and the Node.js program were run in the terminal and the terminal output was used for verification.

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Run npm init -y	package.json is created.	Message "Wrote to ../package.json" displayed; file contains name, version and scripts.	Pass
TC-02	Run npm install lodash	Lodash is installed and listed under "dependencies" in package.json.	Package added; "lodash" appears under dependencies; 0 vulnerabilities reported.	Pass
TC-03	Run node app.js (default numbers 12, 7, 45, 3, 28, 19)	Sum 114, Maximum 45, Minimum 3, Average 19.	Sum: 114, Maximum: 45, Minimum: 3, Average: 19.	Pass
TC-04	Run node app.js -8 15 0 -3 (negative numbers)	Sum 4, Maximum 15, Minimum -8, Average 1.	Sum: 4, Maximum: 15, Minimum: -8, Average: 1.	Pass
TC-05	Run node app.js 42 (single number, boundary case)	Sum, Maximum, Minimum and Average are all 42.	All four values displayed as 42.	Pass
TC-06	Run node app.js abc (invalid input)	Message "Invalid input: please enter numbers only." and the program stops.	Message displayed; program ended with exit code 1.	Pass

9 Output

Output 1: NPM project initialization, package installation and program execution

```
Terminal — student@vm: ~/Q7-NPM-and-package-json

student@vm:~/Q7-NPM-and-package-json$ npm init -y
Wrote to /home/student/Q7-NPM-and-package-json/package.json:

{
  "name": "q7-npm-and-package-json",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "devDependencies": {},
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

student@vm:~/Q7-NPM-and-package-json$ npm install lodash

added 1 package, and audited 2 packages in 2s

found 0 vulnerabilities

student@vm:~/Q7-NPM-and-package-json$ node app.js
Numbers : [ 12, 7, 45, 3, 28, 19 ]
Sum      : 114
Maximum  : 45
Minimum  : 3
Average  : 19

student@vm:~/Q7-NPM-and-package-json$ node app.js -8 15 0 -3
Numbers : [ -8, 15, 0, -3 ]
Sum      : 4
Maximum  : 15
Minimum  : -8
Average  : 1

student@vm:~/Q7-NPM-and-package-json$ █
```

Observed execution results

Item	Observed Value
Project initialization	package.json created successfully.
External package	lodash installed successfully.
Installation result	1 package added; 2 packages audited; 0 vulnerabilities found.
Numbers (default)	[12, 7, 45, 3, 28, 19]
Sum / Maximum / Minimum / Average	114 / 45 / 3 / 19
Numbers (negative test)	[-8, 15, 0, -3] → Sum 4, Maximum 15, Minimum -8, Average 1

10 Result

NPM and package.json were successfully demonstrated. The Lodash external package was installed and used in a Node.js application to calculate the sum, maximum, minimum and average of a list of numbers. The application was executed and the output was verified in the terminal.

End of Observation Record